

Experiment No : 2

Title : Perform recursive Function binary search using C.

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Batch: B

Program:

```
#include <stdio.h>
```

```
int binarysearch(int arr[], int l, int r, int target)
```

```
{
```

```
    if (l > r)
```

```
        return -1;
```

```
    int mid = l + (r - l) / 2;
```

```
    if (arr[mid] == target)
```

```
        return mid;
```

```
    else if (arr[mid] > target)
```

```
        return binarysearch(arr, l, mid - 1, target);
```

```
    else
```

```
        return binarysearch(arr, mid + 1, r, target);
```

```
}
```

```
int main()
```

```
{
```

```
    int arr[] = {1, 2, 3, 4, 5};
```

```
    int n = sizeof(arr) / sizeof(arr[0]);
```

```
    int target, z;
```

```
    printf("Enter target: ");
```

```
    scanf("%d", &target);
```

```
    z = binarysearch(arr, 0, n - 1, target);
```

```
    if (z != -1)
        printf("%d is present at index %d", target, z);
    else
        printf("%d is not present", target);

    return 0;
}5
```

Output:

Output

```
Enter target: 5
5 is present at index 4
```

```
=== Code Execution Successful ===
```